Joshua Yuen

Boston, MA | (302) 688 9416 | yuen.jo@northeastern.edu | LinkedIn

EDUCATION

Northeastern University, Boston, MA— *Candidate for BS in Electrical Engineering and Physics* | *May '29 (Expected)*
- GPA: 3.97/4.00 | Dean's List (All Semesters) | National Merit Scholar (Sponsored by Northeastern University)
- Awarded Gerald F. Tonks Endowed Scholarship (2025-26)
- John Martinson Honors Student: Awarded Self-Directed Learner, Impact, and Community Engagement Badges
- Engineering Honors Society Tau Beta Pi & Electrical & Computer Engineering Honors Society IEEE-Eta Kappa Nu
- Member of NU Robotics's Marine Robotics Team

RELEVANT COURSEWORK

- Fdmt. Linear Systems, Fdmt. Electronics, Fdmt. Electromagnetics, Thermodynamics and Statistical Mechanics, Quantum Mechanics, Differential Equations and Linear Algebra, Calculus III

SKILLS

- **Hardware**: KiCad, Embedded Design, Raspberry Pi, Optical Systems/Interferometry, Linear Systems/Control Theory, Soldering & Circuit Assembly, Microcontrollers,
- **Software**: MATLAB, LaTeX, C++ (basic), Python (basic)

RELEVANT EXPERIENCE

Microscale Acoustic and Photonic Systems (MAPS) Lab, Boston, MA— *Undergrad Researcher* | Nov '25 - Present
- Develop single shot Michelson Interferometer for characterization of microelectromechanical systems (MEMS)
- Designed and assembled $6,000 magnification system for MEMS devices in a cryogenic vacuum probe station
- Updated program calculating displacement of MEMS in a Michelson Interferometer under quadrature condition using the signal-to-noise ratio calculated using experimental parameters from spectrum analyzer and oscilloscope

Embedded Design Course Assistant, Boston, MA— *Hourly Course Assistant (HCA)* | Sep '26 - Present
- Sharpening communication skills and independence via office hours with no professor present; developing a framework for effective communication to facilitate learning in students.

NEU Satellite Labs, Boston, MA— *Electrical Team* | Dec '25 - May '26; *Solar Array Electrical Lead* | Sep '25 - May '26
- Led as second-in-command under Electrical Team Lead to coordinate and assign electrical projects, and instruct new members through the creation of onboarding curriculum

Simply Code Nonprofit, Philadelphia, PA— *Chief Technology Officer* | Aug '24 - Feb '25
- Designed infrastructure to support operations including building and launching the organization's 1st website
- Coordinated volunteers and led the Technology Department, overseeing the planning, execution, and delivery of key technology projects, including partnerships with other organizations in the region

Drexel University Materials Science Summer Institute, Philadelphia, PA–*Lab Student* | Jul '23
- Electrospun carbon nanofibers to study how varying electric field strength during creation process affected fiber diameter; analyzed fibers using scanning electron microscopy to determine relationship between variables

PROJECTS & PUBLICATIONS

Personal Project: IMU-Based Hand Gesture Interface for 3D Design and Modeling | Aug '25 - Present
- Develop glove-based input system using 9-axis IMU sensors to enable real-time hand gesture control and object manipulation in Blender 3D creation software
- Design the software-hardware pipeline for motion tracking, gesture interpretation, and CAD integration
- Research shape memory alloy-based haptic feedback to provide tactile responses to virtual object interactions

Class Project: Embedded Programming–Spider Robot | May '26 - Jul '26
- Uploaded C++ to spider robot FPGA to allow for control of movement; integrated lidar to allow for obstacle detection

Class Project: Occlusion Filtering in License Plate Recognition | May '26 - Jul '26
- Developed Optical Character Recognition Model using OpenCV (pyTesseract) to read occluded license plate images.
- Observed 49% increase in detected license plates and 113% increase in successful license plate readings

Class Project: Analog High, Low, and Bandpass Filter Electrocardiogram | Jan '26 - May '26
- Integrated first-order discrete component filters into a one instrument to measure heart rate; accurate within 1 bpm

NEU Satellite Lab Club Project: Deployable Solar Array for Small Satellites | Sep '25 - May '26
- Investigated power budgets and researched possible circuit schematics for integrating sensing arrays
- Awarded PEAK Base Camp Grant from Northeastern University to further project development

Publication: *Hide and Seek: A Multimodal Approach to Human Detection* | Jul '24
- Co-authored research paper under faculty mentorship; investigated the results of integrating IR sensing, motion profiling, and mmWave sensing into facial recognition software to increase accuracy in detecting occluded faces and darker skin tones. Implemented a machine learning model to interpret results
- Published in the 39th Volume of the Journal of the Pennsylvania Governor's School for the Sciences